

Fundamental Partial Differential Equations

A Unified Study of Diffusion, Wave Propagation, and Convection

Abstract

This report introduces key classes of partial differential equations, namely the heat, wave, and convection equations. These models illustrate three fundamental physical mechanisms: diffusion, oscillation, and transport. Their nonlinear extensions and the role of viscosity are discussed, culminating in Burgers' equation as a unifying model.

1. Introduction

Partial differential equations (PDEs) describe the evolution of physical systems in both space and time:

$$u = u(x, t)$$

Three fundamental processes arise repeatedly:

- Diffusion (smoothing)
- Wave propagation (oscillation)
- Convection (transport)

2. Heat Equation (Diffusion)

Governing equation

$$\frac{\partial u}{\partial t} = \nu \frac{\partial^2 u}{\partial x^2}$$

Properties

- Smooths out irregularities
- Reduces gradients over time
- Infinite propagation speed (idealised model)

3. Wave Equation

Governing equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

Solution

$$u(x, t) = f(x - ct) + g(x + ct)$$

Properties

- Propagation in two directions
- Shape preserved
- Oscillatory behaviour

4. Linear Convection

Governing equation

$$\frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} = 0$$

Solution

$$u(x, t) = f(x - ct)$$

Properties

- Pure transport
- No distortion
- One-directional motion

5. Nonlinear Convection

Governing equation

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = 0$$

Key features

- Speed depends on u
- Solution distorts over time
- Characteristics may intersect

Shock formation

- Gradients steepen
- Discontinuities form
- Classical solutions break down

6. Burgers' Equation

Governing equation

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = \nu \frac{\partial^2 u}{\partial x^2}$$

Interpretation

Combines:

- Nonlinear steepening
- Diffusive smoothing

Behaviour

- Smooth transition layers instead of shocks
- Balance between transport and dissipation

7. Numerical Considerations

Convection (CFL condition)

$$\frac{\max |u| \Delta t}{\Delta x} \leq 1$$

Diffusion condition

$$\frac{\nu \Delta t}{\Delta x^2} \leq \frac{1}{2}$$

8. Unified Perspective

Equation	Primary Effect	Shape	Key Behaviour
Heat	Diffusion	Smooths	Decay of gradients
Wave	Oscillation	Preserved	Bidirectional motion
Linear convection	Transport	Preserved	Translation
Nonlinear convection	Transport	Distorts	Shock formation
Burgers'	Mixed	Partially smooth	Regularised shocks

9. Visual Comparison

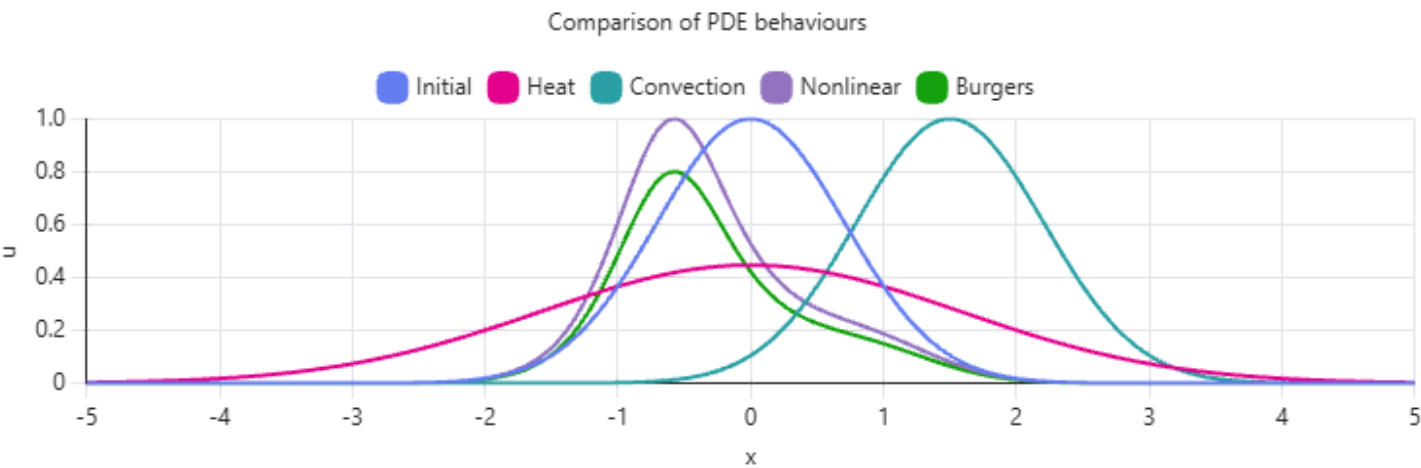


Figure 1: Comparison of PDE solutions for a Gaussian initial condition. Diffusion spreads the profile, convection translates it, nonlinear convection introduces steepening, and Burgers' equation smooths shock formation

10. Discussion

These equations form a conceptual hierarchy:

1. Linear convection: simplest transport
2. Nonlinear convection: introduces self-interaction
3. Burgers' equation: adds physical realism through diffusion

The heat and wave equations provide complementary behaviours:

- diffusion removes structure
- wave propagation preserves it

11. Conclusion

This unified framework highlights:

- Distinct physical mechanisms across PDE classes
- The role of nonlinearity in generating shocks
- The importance of diffusion in stabilising solutions

These ideas underpin much of modern applied mathematics, fluid dynamics, and numerical analysis.

References

1. Evans, L. C., *Partial Differential Equations*
2. LeVeque, R. J., *Finite Volume Methods for Hyperbolic Problems*